THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 8: **Introduction to Trigonometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $\tan A = \frac{4}{3}$, then $\sin A$ is: 1

(a) $\frac{3}{5}$ **(b)** $\frac{4}{5}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{5}{4}$

2. The value of $\frac{2\tan 30^\circ}{1-\tan^2 30^\circ}$ is: 1

(a) $\sqrt{3}$ **(b)** $\frac{1}{\sqrt{3}}$ **(c)** 1 **(d)** 0

3. The value of $\cos 0^\circ + \sin 90^\circ + \tan 45^\circ$ is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 0

4. If $\sec A = 2$ (A acute), then A is: 1

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

5. A student states: "$\sin\theta = \frac{4}{3}$ for some acute angle θ." The correct response is: 1

(a) Correct

(b) Incorrect; $\sin\theta \le 1$ since hypotenuse is the longest side

(c) Correct for $\theta > 45^\circ$

(d) Cannot be determined

6. $(1+\tan^2 A)\cos^2 A$ equals: 1

(a) 0 **(b)** 1 **(c)** $\sin^2 A$ **(d)** $\tan^2 A$

7. For $A = 30^\circ$, Student A says $\sin 2A = 2\sin A = 1$; Student B says $\sin 2A = \sin 60^\circ = \frac{\sqrt{3}}{2}$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. If $\cos A = \frac{12}{13}$, then $\tan A$ is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{5}{13}$ **(d)** $\frac{13}{12}$

9. $\frac{1-\tan^2 45^\circ}{1+\tan^2 45^\circ}$ equals: 1

(a) $\tan 90^\circ$ **(b)** 1 **(c)** $\sin 45^\circ$ **(d)** 0

10. $\sec^2\theta - 1$ is equal to: 1

(a) $\sin^2\theta$ **(b)** $\cos^2\theta$ **(c)** $\tan^2\theta$ **(d)** $\cot^2\theta$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $\cot\theta = \frac{7}{8}$, evaluate $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$. 2

12. Evaluate: $\frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$. 2

13. If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$, $0^\circ, find A and B$. 2

14. A student claims that $\sin(A+B) = \sin A + \sin B$. Check the claim for $A = B = 30^\circ$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Determine 3

(i) $\sin A, \cos A$

(ii) $\sin C, \cos C$.

16. Prove that $(\csc\theta - \cot\theta)^2 = \frac{1-\cos\theta}{1+\cos\theta}$. 3

17. Prove that $\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A} = 2\sec A$. 3

OR

Prove that $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$. 3

18. If $3\cot A = 4$, check whether $\frac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$. 3

OR

In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Determine $\sin P$, $\cos P$ and $\tan P$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\csc\theta$. 5

OR

Prove that $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$. 5

20. **Case study:** A wheelchair ramp is in the shape of a right triangle with horizontal base 12 m and vertical rise 5 m. Let θ be the angle of the ramp with the ground. 5

(i) Find the length of the ramp.

(ii) Find $\sin\theta$, $\cos\theta$ and $\tan\theta$.

(iii) Verify $\sin^2\theta + \cos^2\theta = 1$.

(iv) A student says $\tan\theta > 1$ for this ramp. Comment.

OR

If $\sec\theta + \tan\theta = p$, show that $\sin\theta = \frac{p^2-1}{p^2+1}$. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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